35 Edge AI + IoT Mini Projects (2 Weeks)

□ A. Industrial Monitoring (1–10)

1. Motor Temperature Anomaly Detection

Simulate motor temperature with noise and detect overheating conditions using threshold or simple ML.

2. Vibration-Based Machine Fault Detection

Generate vibration signals (sinusoidal + noise) and detect abnormal spikes indicating faults.

3. Pump Health Monitoring System

Simulate pump behavior (vibration + temperature) and detect failure patterns.

4. Bearing Failure Prediction

Simulate increasing vibration trend over time and detect early failure conditions.

5. Industrial Noise Level Monitoring

Simulate sound levels and detect abnormal noise events.

6. Conveyor Belt Fault Detection

Simulate periodic vibration patterns and detect irregularities.

7. Machine Runtime Monitoring

Track simulated ON/OFF cycles and detect abnormal operation durations.

8. Compressor Pressure Monitoring

Simulate pressure fluctuations and detect abnormal pressure spikes.

9. Industrial Current Monitoring

Simulate current usage and detect overload conditions.

10. Equipment Shock Detection

Detect sudden shocks or impacts using simulated spikes.

□ B. Environmental & Smart Systems (11–20)

11. Room Temperature Anomaly Detection

Simulate room temperature cycles and detect overheating.

12. Smart Humidity Monitoring

Simulate humidity variations and detect abnormal conditions.

13. Air Quality Monitoring System

Simulate air quality index and classify good/bad conditions.

14. Water Level Monitoring System

Simulate tank level rise/fall and detect overflow/empty states.

15. Smart Irrigation Monitoring

Simulate soil moisture and trigger irrigation alerts.

16. Fire/Smoke Detection System

Simulate gas sensor readings and detect abnormal spikes.

17. Smart Lighting Control

Simulate ambient light and adjust lighting conditions.

18. Indoor Comfort Monitoring

Combine temperature + humidity and classify comfort levels.

19. Leak Detection System

Simulate water flow anomalies and detect leaks.

20. Weather Pattern Monitoring

Simulate multiple environmental variables and detect abnormal weather patterns.

⚡ C. Energy & Power Monitoring (21–25)

21. Smart Energy Consumption Monitoring

Simulate power usage and detect unusual consumption spikes.

22. Peak Load Detection System

Identify peak energy usage periods.

23. Voltage Fluctuation Monitoring

Simulate voltage variations and detect instability.

24. Solar Power Output Monitoring

Simulate solar generation and detect drops in output.

25. Battery Health Monitoring

Simulate battery charge/discharge cycles and detect anomalies.

□ D. Industrial Networks & Data Monitoring (26–30)

26. MQTT Data Anomaly Detection

Analyze incoming MQTT data and detect irregular patterns.

27. Modbus Data Monitoring (Simulated)

Simulate Modbus-like data and detect abnormal values.

28. Network Traffic Monitoring

Simulate data traffic and detect unusual spikes.

29. Multi-Sensor Data Fusion

Combine 2–3 sensors and detect inconsistencies.

30. Data Loss Detection System

Detect missing or delayed MQTT messages.

□ E. Edge AI / Advanced (31–35)

31. Motion Detection System (Simulated)

Simulate motion events and detect unusual activity.

32. Sound-Based Event Detection

Simulate sound intensity and detect abnormal events.

33. Predictive Temperature Model

Predict next temperature value using simple regression.

34. Appliance Usage Monitoring

Simulate usage patterns and detect unusual behavior.

35. Multi-Parameter Anomaly Detection

Combine multiple variables and detect complex anomalies.

□ Instructions for Students (Attach This)

Each project must include:

- Data simulation (realistic patterns + anomalies)
- Edge processing (Python)
- AI logic (threshold / simple ML)
- MQTT communication
- Node-RED dashboard

- Code on GitHub